A13485W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6. CONDENSED-MATTER PHYSICS

MICHAELMAS TERM 2020

Tuesday, 13 October

Opening time: 09.30 am (BST)

You have two hours and thirty minutes to complete the paper and upload your answer file

Answer **two** *questions.*

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors is available at the Physics website.

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) In the face-centered cubic (FCC) lattice, for each site with fractional coordinates $[x, y, z]$ there are identical sites at coordinates $[x+\frac{1}{2}, y+\frac{1}{2}, z]$, $[x+\frac{1}{2}, y, z+\frac{1}{2}]$, and $[x, y+\frac{1}{2}, z+\frac{1}{2}]$. Write the contribution to the structure factor of four atoms located on sites related in this way, and show that Bragg peaks are extinct unless the Miller indices h, k and l are either all even or all odd. Show that, for non-extinct Bragg peaks, the contribution to the structure factor for each atom in the primitive basis is given by

$$4f\mathrm{e}^{2\pi\mathrm{i}(hy+ky+lz)},$$

where f is the atomic scattering factor (X-rays) or Fermi length (neutrons). [5]

(b) Silicon has a FCC lattice with a 2-atom basis (8 atoms in the conventional unit cell), with coordinates $[0, 0, 0]$ and $[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$. Show that, for silicon, the square of the structure factor for non-extinct Bragg peaks is given by:

$$|S_{\{hkl\}}|^2 = 32 f_{\mathrm{Si}}^2 \left(1 + \cos\frac{n\pi}{2}\right),$$

where n is an integer given by a combination of h, k and l, which you should determine. [6]

(c) The lattice constant of silicon is $a = 0.543\,\mathrm{nm}$.

(i) Write the expression of the spacing between atomic planes $d_{(hkl)}$ in terms of the Miller indices and the lattice constant.

(ii) Show that the powder diffraction multiplicity of the (002) reflection is $m_{(002)} = 6$, and that of the (111) reflection is $m_{(111)} = 8$.

[4]

(d) A neutron diffraction experiment is performed on a powder sample of silicon, using a neutron wavelength of 0.1542 nm. Complete the following table, having calculated the values of $d_{\{hkl\}}$, $2\theta_{\{hkl\}}$, the multiplicity m and the product $|S_{\{hkl\}}|^2 \times m_{\{hkl\}}$ for the listed peaks. The measured intensities (last column) are in arbitrary units, and have been corrected with the geometrical factors of the diffractometer. Comment on the level of agreement you have obtained between I_{calc} and I_{meas}.

peak	$\{hkl\}$	$d_{\{hkl\}}$	$2\theta_{\{hkl\}}$	m	$I_{\mathrm{calc}} = \lvert S_{\{hkl\}}\rvert^2 \times m_{\{hkl\}}$	I_{meas} (Measured intensity)
a	111					1000
b	200					0
c	220					2755
d	311					2619
e	400					1203

[7]

(e) An equivalent experiment is performed with X-rays having the same wavelength. (i) How should the calculated structure factors be modified? (ii) How does the ratio $I_{\mathrm{X}}/I_{\mathrm{n}}$ between the intensities measured with X-rays and neutrons for the same Bragg peak evolve as a function of $2\theta_{hkl}$? [3]

2. (a) In the Debye theory of atomic vibrations, the Debye wavevector $k_{\rm D}$ is calculated by making the number of quantum mechanical oscillator modes equal to the number of degrees of freedom of a crystal with N atoms. Show that the Debye wavevector in the case of a 2-dimensional crystal and 1 atom per unit cell is given by:

$$k_{\rm D} = a\sqrt{n},$$

where a is a numerical constant to be determined and $n = N/A$ is the area density of atoms. [5]

(b) Using the result of part (a), find expressions for the Debye (cut-off) angular frequency $\omega_{\rm D}$, the Debye temperature $T_{\rm D}$ and the density of phonon states $g(\omega)$ for

(i) an isotropic acoustic phonon mode in 2 dimensions with two degenerate polarisations and a *linear* dispersion relation $\omega = ck$, where c is the sound velocity, and

(ii) an isotropic phonon mode in 2 dimensions with a single polarisation and *quadratic* dispersion relation $\omega = Dk^2$, where D is known as the stiffness constant.

In particular, show that in case (i) $g(\omega) = \gamma\omega$ and that in case (ii) $g(\omega) = \gamma'$, and find constants γ and γ'. [6]

(c) Using the results from part (b), calculate the molar specific heat for the two cases in the low-temperature approximation, and show that it has the general expression:

$$c_v = R\,\alpha \left(\frac{T}{T_{\rm D}}\right)^{\beta},$$

where R is the molar gas constant, while α and β are dimensionless numerical constants that you should determine in the two cases.
[The following integrals are useful: $\int_0^\infty x/(\mathrm{e}^x - 1)\mathrm{d}x = \pi^2/6$ and $\int_0^\infty x^2/(\mathrm{e}^x - 1)\mathrm{d}x = 2\zeta(3)$, where ζ is the Riemann zeta function and $\zeta(3) \approx 1.202$.] [6]

(d) Graphene is a two-dimensional form of carbon (atomic weight 12.01 $\mathrm{g\,mol^{-1}}$), having a mass per unit area of 0.38×10^{-3} $\mathrm{g\,m^{-2}}$. In-plane displacements of carbon atoms give rise to transverse and longitudinal acoustic phonons (which you can consider as degenerate in the remainder), having a linear dispersion relation as in (b)(i), with an average velocity of sound $c = 16,800\ \mathrm{m\,s^{-1}}$. Out-of-plane carbon displacements give rise to phonons dispersing quadratically as in (b)(ii), with a stiffness constant $D = 6.06 \times 10^{-7}\ \mathrm{m^2\,s^{-1}}$.

Assuming that graphene had one atom per unit cell, show that the specific heat of graphene in the low-temperature limit is given by the expression:

$$c_v = R\,\left(c_1 T + c_2 T^2\right)$$

and calculate the values of c_1 and c_2. [5]

(e) Calculate the specific heat of graphene at room temperature, compare with the experimental value of $9\ \mathrm{J\,K^{-1}\,mol^{-1}}$, and explain the discrepancy. Is the low-temperature approximation justified in this case? Does the fact that graphene has two atoms per unit cell make any difference? [3]

3. (a) The number of thermally excited electrons $n(T)$ and holes $p(T)$ per unit volume in a semiconductor can be approximated as:

$$n(T) = \frac{1}{4}\left(\frac{2m_{\rm e}^*}{\beta\pi\hbar^2}\right)^{3/2} {\rm e}^{-\beta(\epsilon_{\rm c}-\mu)}\ ;\ p(T) = \frac{1}{4}\left(\frac{2m_{\rm h}^*}{\beta\pi\hbar^2}\right)^{3/2} {\rm e}^{-\beta(\mu-\epsilon_{\rm v})}$$

where $\beta = 1/k_{\rm B}T$.

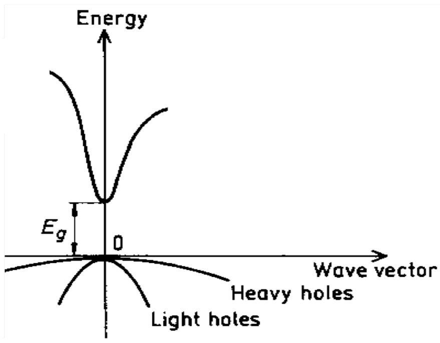
(i) Define the quantities μ, $m_{\rm e}^*$, $m_{\rm h}^*$, $\epsilon_{\rm c}$ and $\epsilon_{\rm v}$ and write down the expression for the energy gap $E_{\rm g}$.

(ii) Discuss the importance of approximating the valence and conduction bands as parabolic, and

(iii) explain why the Boltzmann statistics can be employed in the derivation instead of the Fermi-Dirac statistics for light doping if $\epsilon_{\rm c} - \epsilon_{\rm v} \approx 1.5$ eVand T=300 K.

[6]

(b) The band structure of semiconductor gallium arsenide (GaAs) near the conduction band minimum/valence band maximum is sketched in the figure below.



GaAs has two valence bands, known as light holes (lh) and heavy holes (hh), with $m_{\rm lh}^* = 0.082\ m_0$ and $m_{\rm hh}^* = 0.45\ m_0$, and a single conduction band with $m_{\rm e}^*$=0.067 m_0, where m_0 is the free electron mass. The GaAs energy gap is $E_{\rm g} = 1.42$ eV. Assuming that the formulas given above apply for the two valence bands and that $\epsilon_{\rm v}$ is the same for both the light and heavy holes, calculate the ratio of the numbers of heavy to light holes and show that it does not depend on temperature. [3]

(c) Show that the total number of thermally excited holes in undoped GaAs is given by the formula:

$$p_{\rm tot}(T) = \frac{1}{4}\left(\frac{2m_p^*}{\beta\pi\hbar^2}\right)^{3/2} {\rm e}^{-\beta(\mu-\epsilon_{\rm v})}$$

where m_p^* must be calculated in terms of $m_{\rm lh}^*$ and $m_{\rm hh}^*$. [4]

(d) Derive the law of mass action for GaAs and calculate the density of carriers for undoped GaAs at T= 300 K and T= 150 K. Assuming that the law of mass action remains valid, write the general expression of $n(T)$ for doped GaAs and calculate the density of electrons and holes at the same two temperatures T= 300 K and T= 150 K for p-doped GaAs with an acceptor density of 1×10^{11} m^{-3}. [8]

(e) Propose an experiment to measure the carrier density as a function of temperature, and explain whether it is more sensitive to electrons or to holes. [4]

4. A simple one-dimensional model of a ferromagnet consists of an infinite linear chain of atoms interacting with the classical Hamiltonian

$$H = -J \sum_{<i,j>} \vec{S}_i \cdot \vec{S}_j - \kappa \sum_i (S_i^z)^2$$

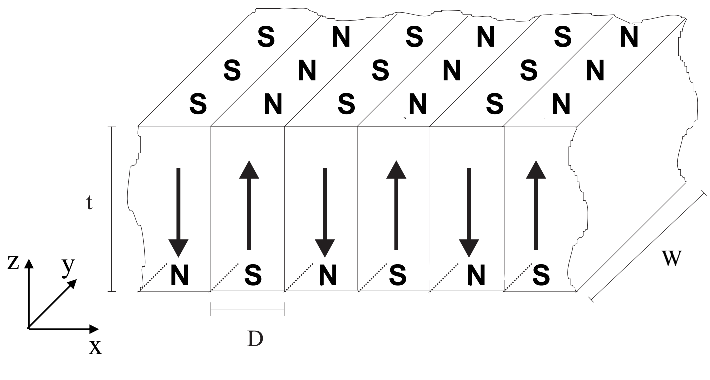
where $J > 0$, $\kappa > 0$ and the first summation runs over nearest neighbours. A model of a magnetic domain wall consists of spins that uniformly rotate over a length L of this chain, so that spins at the origin point horizontally, spins at $x = +L/2$ point vertically up and spins at $x = -L/2$ point vertically down.

(a) Discuss the terms in the Hamiltonian. Draw a chain with lattice constant a as described above, and explain why spins very far from the domain boundary should be ferromagnetically aligned. [3]

(b) Write the expression for the average energy of the domain boundary as a function of the length L, subtracting the energy of a uniform ferromagnetic chain from that of the chain described above. [4]

(c) With the assumption $L \gg a$, show that the length of the chain that minimises the energy is $L = c_1 a\sqrt{J/\kappa}$, where c_1 is a constant you should determine, and find the minimum energy of the chain as a function of J and κ. [The following limit expression may be useful: $\sum_{m=1}^{\alpha/\delta} f(m\delta) \approx (1/\delta) \int_0^\alpha f(x)\mathrm{d}x$ when $\delta \ll \alpha$.] [6]

(d) A more realistic model for a domain wall is a 2D lattice of such chains with lattice constant a in both the y and z directions. (i) find the total energy of a single domain wall of this kind, extending a height t along z and a width W along y. (ii) Find the total energy per unit volume of a system of parallel domain walls, regularly spaced by a distance D along x (see figure below). [6]



(e) Explain why for real magnets it may be favourable to form domain walls in spite of the domain wall energy always being positive and describe the role of the magnetostatic energy. In the limit $t \ll D < W$, the magnetostatic energy per unit volume can be approximated by $E_{\mathrm{mag}} \approx c_2\, \mu_0 M^2 D/t$ where c_2 is a numerical constant and M is the magnetisation. Show that at equilibrium $D \propto t^\beta$, and determine β. [6]